

Arithmetic Foundation: Decimals, Place Value & Operations Exam (Set 5)

Name:

Class:

Student ID:

Time: 30 Mins

Total: 100 Pts

Score:

Section 1: Place Value & Identification (4 pts each, 20 pts total)

Score: _____

1. For the number 348.529:

- (a) Which digit is in the **tens** place? _____ (b) Which digit is in the **tenths** place? _____
(c) Which digit is in the **hundredths** place? _____ (d) What is the place value represented by the digit 9?
? _____

2. [Leading Zeros & Significant Figures] State the number of significant figures in each value:

(a) 0.00397 has _____ sig figs

(b) 50.04 has _____ sig figs

(c) 0.075 has _____ sig figs

(d) 6.00 has _____ sig figs

Section 2: Decimal Addition, Subtraction & Alignment (3 pts each, 24 pts total)

Score: _____

Line up the decimal points vertically before calculating. Do not guess!

(a) $0.9 + 0.1 =$ _____(b) $3.95 + 0.05 =$ _____(c) $12.8 + 4.35 =$ _____(d) $0.07 + 0.93 =$ _____(e) $1 - 0.25 =$ _____(f) $5 - 0.08 =$ _____(g) $10.4 - 3.85 =$ _____(h) $0.5 - 0.05 =$ _____**Section 3: Decimal Multiplication, Division & Shift Rules (4 pts each, 28 pts total)**

Score: _____

3. Mental Sprint: Move the decimal point according to division and multiplication rules:

(a) $5.4 \div 0.6 = 54 \div 6 =$ _____

(b) $30 \div 0.5 = 30 \times 2 =$ _____

(c) $0.36 \div 0.09 = 36 \div 9 =$ _____

(d) $4.8 \div 0.12 = 480 \div 12 =$ _____

(e) $2.5 \times 0.04 =$ _____

(f) $1.2 \times 0.5 =$ _____

(g) $40 \div 0.2 =$ _____

Section 4: Decimals, Fractions & Percentages Conversions (4 pts each, 28 pts Score: total) _____

4. Convert each decimal to a percentage (Multiply by 100%):

(a) $0.075 \rightarrow$ _____ %

(b) $0.75 \rightarrow$ _____ %

(c) $0.008 \rightarrow$ _____ %

(d) $1.25 \rightarrow$ _____ %

5. [Rounding & Precision Limits] A length is measured as 6.0 cm rounded to 1 decimal place.

- (a) Explain why writing 6 cm is NOT the same as writing 6.0 cm in terms of measurement precision.
(b) What is the lower bound and upper bound for the true value of this length?

(a) Explanation:

(b) Lower bound = _____ cm; Upper bound = _____ cm

6. [Division with Zero in Quotient] Solve $40 \div 42$ to 2 decimal places using standard long division.
Show all steps:

Show your long division here :

$40 \div 42 \approx$ _____



[Foundational Clinic 1] Answer Key & Scoring Guide

Section 1: Place Value & Identification (20 pts)

1. (a) 4; (b) 5; (c) 2; (d) Thousandths ($\frac{9}{1000} = 0.009$).
2. (a) 3 (3, 9, 7); (b) 4 (5, 0, 0, 4); (c) 2 (7, 5); (d) 3 (6, 0, 0).

Section 2: Decimal Addition & Subtraction (24 pts)

- (a) 1.0; (b) 4.00; (c) 17.15; (d) 1.00; (e) 0.75; (f) 4.92; (g) 6.55; (h) 0.45.

Section 3: Multiplication & Division Rules (28 pts)

- (a) 9 (⚠ NOT 0.9!); (b) 60; (c) 4; (d) 40; (e) 0.1; (f) 0.6; (g) 200.

Section 4: Decimals & Percentages (28 pts)

4. (a) 7.5% ; (b) 75%; (c) 0.8%; (d) 125%.
5. (a) 6 cm means precision to the nearest whole unit (interval $5.5 \leq x < 6.5$), while 6.0 cm means precision to the nearest tenth (interval $5.95 \leq x < 6.05$). 6.0 is ten times more precise.
(b) Lower bound = 5.95 cm; Upper bound = 6.05 cm.
6. Long division: $40.000 \div 42 = 0.952... \approx 0.95$.